

4 Phloem sap is transported within phloem sieve tubes.

- (a) Monosaccharides and disaccharides are transported as part of phloem sap, but polysaccharides are not components of phloem sap.

Define the terms disaccharide **and** polysaccharide.

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..... [2]

- (b) Sieve tubes are composed of phloem sieve tube elements.

Fig. 4.1 is a diagram of one sieve tube element and its companion cell.

Some of the structural features of the sieve tube element have not been included in the diagram.

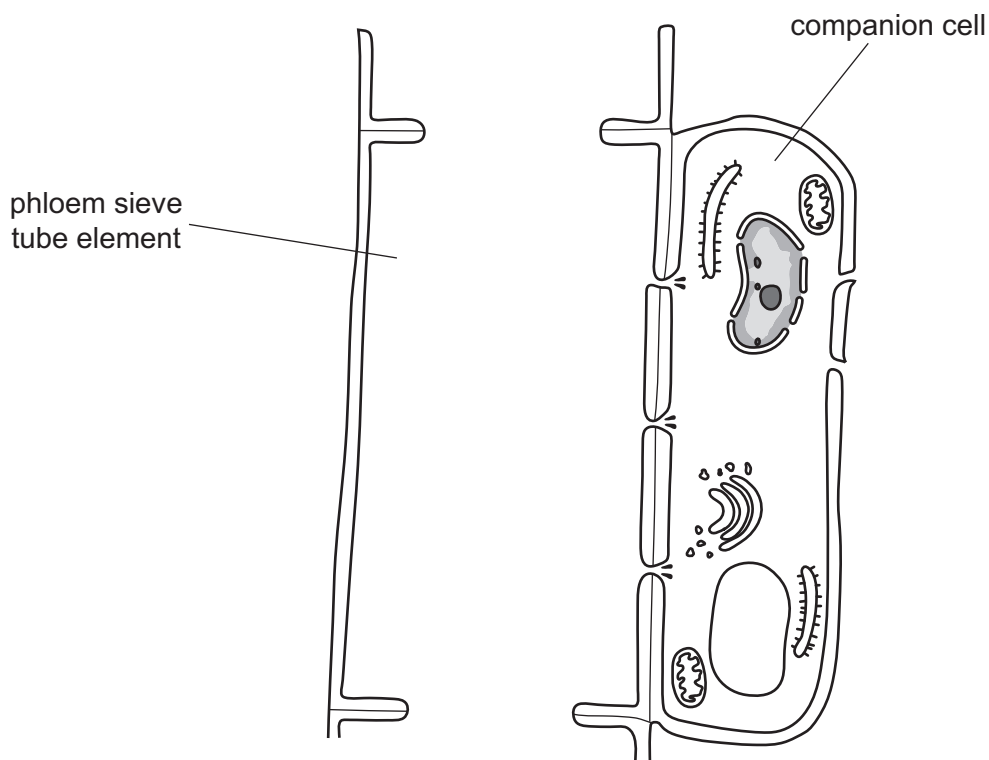


Fig. 4.1

- (i) Complete Fig. 4.1 by drawing **and** labelling the structural features of the sieve tube element that have not been included in the diagram. [3]

- (ii) A pressure gradient in a sieve tube causes the mass flow of phloem sap from the source to the sink.

At the source, a decrease in water potential in the phloem sap and an increase in the hydrostatic pressure of the phloem sap can be measured.

Describe the events that cause each of these changes in the phloem sap.

decrease in water potential

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increase in hydrostatic pressure

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..... [2]

[Total: 7]